

9 Calculus

What students need to learn	Notes
A Differentiation and integration of sums of multiples of powers of x (excluding integration of $\frac{1}{x}$), $\sin ax$, $\cos ax$, e^{ax}	No formal proofs of the results for ax^n , $\sin ax$, $\cos ax$ and e^{ax} will be required.
B Differentiation of a product, quotient and simple cases of a function of a function	
C Applications to simple linear kinematics and to determination of areas and volumes	Understanding how displacement, velocity and acceleration are related using calculus. The volumes will be obtained only by revolution about the coordinate axes.
D Stationary points and turning points	
E Maxima and minima	Maxima and minima problems may be set in the context of a practical problem. Justification of maxima and minima will be expected.
F The equations of tangents and normals to the curve $y = f(x)$	$f(x)$ may be any function which the students are expected to be able to differentiate.
G Application of calculus to rates of change and connected rates of change	The emphasis will be on simple examples to test principles. A knowledge of $dy \approx \frac{dy}{dx} dx$ for small dx is expected.